

Investigating Intraocular Pressure (IOP) Variability in Primary Open-Angle Glaucoma (POAG)

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Abstract:

Background: Primary open-angle glaucoma (POAG) is a leading cause of irreversible vision loss, with intraocular pressure (IOP) recognized as the most significant modifiable risk factor. However, the roles of IOP variability and systemic factors, such as systemic blood pressure, in disease progression remain controversial. This study aims to evaluate how systemic, ocular, and IOP variability metrics interact to predict POAG progression in patients with elevated baseline IOP.

Methods: A longitudinal cohort of 819 participants with elevated IOP (≥ 24 mmHg) but no prior POAG diagnosis or treatment were followed for up to 10 years. Comprehensive demographic, systemic, and ocular variables were assessed. Cox proportional hazards modeling was applied, and covariates were selected using Backward and Forward Selection, then refined with Lasso penalization ($\lambda=0.05$). Violations of the proportional hazards assumption were addressed through stratification. Subgroup analyses using log-rank tests compared progression risk across high- and low-variability groups (defined by the median SD IOP and Range IOP values) and systemic or demographic characteristics.

Results: Baseline and follow-up IOP metrics, including mean follow-up IOP (HR: 1.75, 95% CI: 1.54–1.98) and maximum IOP (HR: 1.60, 95% CI: 1.41–1.81), were significant predictors of POAG progression. IOP variability metrics, SD IOP (HR: 1.35, 95% CI: 1.19–1.52) and Range IOP (HR: 1.40, 95% CI: 1.24–1.58), also showed independent associations with progression. Subgroup analyses revealed that the effects of IOP variability differed significantly across systemic and demographic subgroups. Hypertension was included as a stratification variable in the final Cox model to account for its time-dependent effects, which achieved a mean concordance index of 0.7370 (SD: 0.0523).

Conclusions: This study highlights the multifactorial nature of POAG progression, emphasizing the independent roles of both absolute IOP levels and variability, as well as their interaction with systemic conditions. The findings provide actionable insights for improving risk stratification and guiding personalized treatment strategies, particularly for patients with systemic comorbidities. These results have important implications for refining clinical guidelines and enhancing outcomes through tailored glaucoma management.

1. Introduction

Primary open-angle glaucoma (POAG) represents a leading cause of irreversible vision loss worldwide, characterized by a multifactorial and progressive pathology. Elevated intraocular pressure (IOP) has long been established as the primary modifiable risk factor, yet the role of IOP variability in POAG progression remains contentious. Contradictory evidence from landmark studies such as the Advanced Glaucoma Intervention Study (AGIS) and the Early Manifest Glaucoma Trial (EMGT) has fueled ongoing debate. AGIS highlighted an association between greater IOP fluctuations and accelerated visual field loss, whereas EMGT emphasized mean IOP as the critical determinant, undermining the significance of variability [1,2]. This inconsistency underscores the need for further investigation into whether IOP variability constitutes an independent risk factor or merely reflects sustained elevated IOP levels.

Systemic factors, particularly hypertension, further complicate the interplay between IOP dynamics and glaucoma progression. While studies such as the Rotterdam Study identified systemic blood pressure as a potential contributor to IOP elevation and high-tension glaucoma, the relationship with normal-tension glaucoma remains elusive and controversial [3]. These findings suggest a complex interaction between vascular health and glaucoma pathogenesis, which warrants a more integrated analysis of systemic and ocular determinants of POAG progression.

This study addresses the critical gap in understanding the combined impact of systemic factors, IOP variability, and baseline IOP on the risk of POAG progression. We hypothesize that IOP variability exerts an independent influence on glaucoma progression and that systemic conditions, such as hypertension, modulate this effect. By leveraging advanced survival analysis techniques, this research aims to disentangle these interrelated factors, providing nuanced insights into POAG risk stratification. The findings of this study have the potential to refine clinical guidelines, emphasizing personalized approaches that stabilize IOP variability and address systemic comorbidities, ultimately enhancing outcomes for patients at risk of glaucoma.

2. Materials and Methods

2.1 Study Population and Data Description

This study analyzed data from a longitudinal cohort of 819 participants recruited from the United States. These participants were identified as being at risk of developing POAG due to elevated IOP, defined as having at least one prior measurement of IOP ≥ 24 mmHg. The inclusion criteria ensured that no participant had a diagnosis of POAG or had received any treatment for glaucoma at baseline. Participants were followed for up to 10 years, during which periodic assessments of ocular and systemic health were performed. To comprehensively evaluate the risk of progression to POAG, a variety of demographic, systemic, and ocular variables were included in the analysis. These variables were chosen based on their clinical relevance and potential associations with glaucoma progression, ensuring a robust assessment of risk factors and accounting for potential confounders. Data were collected at baseline and during follow-up visits, capturing both static characteristics and dynamic changes over time.

At baseline, demographic factors included age, recorded in years, sex, categorized as male or female, and ethnicity, classified as White, Black, or Other. Systemic health variables included the presence or absence of hypertension, diabetes, migraine, cardiovascular disease (CVD), and a family history of glaucoma. These systemic factors were included due to their hypothesized roles in vascular regulation and optic nerve health. Baseline ocular measurements encompassed IOP measured in millimeters of mercury (mmHg), corneal thickness measured in micrometers, and the vertical cup-to-disc ratio (VCDR), a structural marker of optic nerve health and potential early glaucomatous changes.

During the follow-up period, intraocular pressure was measured periodically, providing data on both absolute levels and variability over time. Absolute IOP measures included the mean IOP during follow-up (FUP IOP), calculated as the average of all IOP measurements, and the maximum IOP (Max IOP), representing the peak pressure recorded. These metrics reflect sustained and extreme levels of IOP, both of which are critical contributors to glaucoma risk. Variability in IOP was assessed using several metrics: the standard deviation of IOP (SD IOP), which captures the extent of fluctuations; the range of IOP (Range IOP), calculated as the difference between the highest and lowest IOP values; and the coefficient of variation for IOP (CV IOP), a standardized measure of variability relative to the mean.

The primary outcome of the study was progression to POAG, defined as the presence of reproducible visual field abnormalities or optic disc deterioration attributed to glaucoma, as determined by masked certified readers and adjudicated by a masked Endpoint Committee. This outcome was represented as a binary variable, with "Status = 1" indicating progression to POAG and "Status = 0" indicating no progression, with censoring applied at the end of the follow-up period. The time variable in the dataset represented either the time to POAG progression for participants who developed the disease or the duration of follow-up for those who did not.

The variables used in this study were catalogued as follows:

- Categorical Variables: Sex, Familiarity, Diabetes, Hypertension, CVD, Migraine, Status, Ethnicity.
- Numerical Variables: Age, Base IOP, Corneal thickness, VCDR, FUP IOP, SD IOP, MAX IOP, Range IOP, CV IOP, Time.

2.2 Descriptive Analysis

Descriptive statistics were performed to summarize the characteristics of the cohort, with results presented separately for patients who progressed to POAG and those who were censored. For categorical variables, absolute and relative frequencies were calculated to describe the distribution within each category, providing insights into the prevalence of systemic and demographic factors across the groups. For numerical variables, the mean and standard deviation (SD) were used to represent central tendency and variability, while the median and interquartile range (IQR) were calculated to provide robust measures of central tendency and spread, particularly for skewed distributions. This dual approach facilitated a comprehensive comparison between progressors and censored patients, highlighting differences in both categorical and continuous variables that could inform further analyses.

Additionally, a correlation matrix was computed for the IOP-related metrics (Base IOP, FUP IOP, SD IOP, Max IOP, Range IOP, and CV IOP) to evaluate potential multicollinearity among these covariates. High correlations between variables can inflate standard errors and compromise the stability of statistical models, making it critical to address multicollinearity before constructing the Cox proportional hazards model. In cases where strong correlations were identified, a selection process was undertaken to retain the variables that provided the most distinct and clinically relevant

information. This approach ensured the robustness and interpretability of the final model, enabling a more accurate assessment of the relationships between IOP metrics and POAG progression.

2.3 Survival Analysis

Survival analysis was employed to explore the progression to POAG over time and assess the impact of key variables. The analysis used Kaplan-Meier survival estimation, log-rank tests, and Nelson-Aalen cumulative hazard estimation, each providing unique insights into the time-dependent nature of disease progression and the role of various risk factors. Together, these methods formed the foundation for subsequent Cox proportional hazards modeling, enabling a more detailed exploration of how specific covariates influenced POAG risk.

2.3.1 Kaplan-Meier Survival Estimation

The Kaplan-Meier method was utilized to estimate the survival probabilities, representing the proportion of participants who had not yet progressed to POAG at different time points. Median survival times were also calculated to provide a clear reference point for when 50% of participants remained free of POAG. This non-parametric approach allowed for visualizing survival distributions across subgroups based on key covariates, such as hypertension. By generating Kaplan-Meier survival curves for each subgroup, the study facilitated direct comparisons of survival probabilities over time. The confidence intervals for these estimates were included to convey the uncertainty associated with the survival probabilities, particularly for later follow-up times as the number of at-risk participants decreased.

2.3.2 Nelson-Aalen Cumulative Hazard Estimation

The Nelson-Aalen estimator was used to compute the cumulative hazard function, which represents the accumulated risk of POAG progression over time. These plots offered a valuable perspective on how risk unfolded over time.

2.3.3 Log-Rank Test

To statistically evaluate differences in survival distributions between groups, log-rank tests were performed. This method compared the observed and expected counts of POAG progression events at each time point across strata, testing the null hypothesis that survival probabilities were identical between groups. For each categorical variable, the corresponding p-value was computed, with statistical significance defined as $p < 0.05$.

2.4 Cox Proportional Hazards Model

The Cox proportional hazards model was employed as the primary analytical method to explore the relationship between predictor variables and the risk of progression to POAG. This model was chosen for its ability to handle time-to-event data while accounting for censoring and multiple covariates simultaneously. The modeling process was carefully structured to ensure statistical rigor and clinical relevance.

Before constructing the statistical models, all numerical variables were standardized to have a mean of zero and a standard deviation of one. This preprocessing step ensured that variables on different scales were directly comparable, improving the stability and interpretability of the model coefficients. Additionally, categorical variables were converted into binary indicators using OneHotEncode, allowing them to be appropriately incorporated into the statistical models.

2.4.1 Initial Model Construction

An initial model was built that included all baseline and follow-up variables as covariates, with the exception of metrics exhibiting high multicollinearity. This comprehensive approach allowed for the evaluation of all available predictors and provided a foundation for subsequent refinement. The purpose of this step was to identify potential contributors to the outcome while capturing the overall predictive capacity of the dataset. To assess the model's performance, the concordance index was calculated using 10-fold cross-validation.

2.4.2 Variable Selection

To identify the most relevant variables, we employed two complementary approaches: Backward Selection and Forward Selection. Both methods used all continuous variables as covariates and categorical variables that were significant based on the Log-Rank Test. The concordance index was chosen as the evaluation metric to compare the predictive capacity of the features selected by each approach. An intermediate refined model was then constructed by combining the variables identified through both selection strategies, ensuring that the strongest contributors to the outcome were retained.

To further simplify the model, reduce complexity, and enhance interpretability, Lasso penalization was applied to the intermediate model. A penalization coefficient (λ) of 0.05 was used to shrink coefficients of less significant variables toward zero, effectively retaining only the most meaningful predictors. This regularization step prevented overfitting and ensured model robustness. Based on the evaluation of the concordance index, it was determined that a simplified model including only the key variables with non-zero coefficients, identified through penalization, maintained comparable predictive accuracy to the more complex penalized model. Given this equivalence and the advantages of simplicity, the refined model was selected as the final model. This decision prioritizes clinical applicability and interpretability, ensuring the model remains practical for integration into decision-making processes without compromising its predictive utility.

2.4.3 Testing the Proportional Hazards Assumption

The proportional hazards assumption, a key requirement of the Cox model, was tested for all included variables using Schoenfeld residuals. This analysis examines whether the effect of each variable on the outcome is consistent over time. Identifying and addressing violations of this assumption ensures the validity of the model. In cases where the assumption was violated adjustments were made. Specifically, variables that violated the assumption were treated as stratification factors, allowing the baseline hazard to vary across strata without estimating a specific hazard ratio for the stratified variable.

2.4.4 Final Model Development and Validation

The final model was constructed using the selected variables and incorporated stratification where necessary. Residual analyses, including Martingale and deviance residuals, were conducted to confirm the model's goodness of fit and to identify any potential outliers or influential observations.

2.5 Analysis of IOP Variability across Patient Subgroups

Building on the identification of SD IOP and Range IOP as significant predictors of progression to POAG in the Cox proportional hazards model, additional analyses were conducted to explore whether the impact of IOP variability differs across patient subgroups. This investigation aimed to examine potential interactions between IOP variability metrics and systemic or demographic characteristics, using the Log-Rank test to evaluate differences in survival distributions. To stratify the cohort based on IOP variability, the values of SD IOP and Range IOP were divided into high and low groups using the respective medians as thresholds. This binary stratification allowed for the comparison of progression risks across groups with different levels of IOP variability.

Key systemic and demographic variables, including sex, familiarity with glaucoma, diabetes, hypertension, cardiovascular disease (CVD), and migraine, were analyzed in conjunction with IOP variability metrics. For each variable, the cohort was further subdivided into subgroups (e.g., males vs. females, hypertensive vs. non-hypertensive), enabling the evaluation of survival outcomes within these categories. The Log-Rank test was applied to compare survival distributions between patients with high and low variability in SD IOP or Range IOP, stratified by each categorical variable. Statistical significance was determined using a threshold of $p < 0.05$ to identify meaningful differences in survival distributions. The rationale behind this analysis was to uncover heterogeneity in how IOP variability influences POAG progression, considering the potential modulating effects of systemic and demographic factors.

2.6 Statistical Software and Visualization

All statistical analyses and visualizations were performed using Python, utilizing the following libraries and tools:

- Data Manipulation: pandas and numpy for efficient handling and processing of datasets.
- Survival Analysis: lifelines for Kaplan-Meier survival curves, Cox proportional hazards models, and log-rank tests.
- Feature Selection: lifelines penalized regression implementation for variable selection.
- Statistical Tests: scipy.stats for basic statistical tests and log-rank comparisons.
- Visualization: matplotlib for custom plot styling and seaborn for enhanced data visualizations.

3. Results

3.1 Descriptive Statistics

A total of 819 observations were analyzed, with no missing data across the variables. The data presented in Table 1 summarize the distribution of categorical variables in the cohort, divided into those who experienced progression to POAG and those who did not. Among the non-progression group, females accounted for 56%, while males represented 44%; in the progression group, the proportions shifted slightly to 59% and 41%, respectively. A similar pattern emerged for family history of glaucoma, where 65% of non-progressors and 59% of progressors reported no family history, while 35% and 41%, respectively, had a positive family history. For diabetes, the prevalence was relatively low, with 89% of non-progressors and 86% of progressors not affected by the condition. Hypertension showed a more noticeable difference, being present in 35% of the non-progression group and rising to 48% in the progression group. Cardiovascular disease was slightly more common in the progression group, with 8% compared to 6% in non-progressors. Similarly, migraine prevalence was nearly identical across groups, with 89% of non-progressors and 88% of progressors not reporting the condition. Ethnic distribution showed little variation, with White participants constituting 71% of the non-progression group and 73% of the progression group, while Black participants made up 22% in both groups. Individuals of other ethnicities accounted for 7% of the non-progressors and 6% of the progressors. Regarding the study outcome, 34% of participants progressed to POAG, while the remaining 66% were censored during follow-up.

Table 1. Absolute and Relative Frequency of Categorical Variables in Patients Who Progressed to POAG Versus Censored Patients

Variable	Categories	Frequency in Censored Group	Frequency in Progression Group
Sex	Female, Male	302 (56%), 238 (44%)	166 (59%), 113 (41%)
Familiarity	No, Yes	352 (65%), 188 (35%)	166 (59%), 113 (41%)
Diabetes	No, Yes	481 (89%), 59 (11%)	239 (86%), 40 (14%)
Hypertension	No, Yes	350 (65%), 190 (35%)	145 (52%), 134 (48%)
Cardiovascular Disease	No, Yes	509 (94%), 31 (6%)	258 (92%), 21 (8%)
Migraine	No, Yes	478 (89%), 62 (11%)	245 (88%), 34 (12%)
Ethnicity	White, Black, Other	381 (71%), 120 (22%), 39 (7%)	203 (73%), 60 (22%), 16 (6%)

Continuous variables provide additional insight into the cohort's clinical and demographic characteristics (Table 2). Patients who progressed to POAG exhibited higher IOP levels across multiple measures. The mean baseline IOP was 25.48 mmHg (SD: 2.99) in the progression group compared to 24.32 mmHg (SD: 2.70) in the non-progression group, with median values of 25.41 mmHg and 24.40 mmHg, respectively. This trend continued during follow-up, where the mean follow-up IOP was 25.02 mmHg (SD: 3.06) among progressors versus 23.21 mmHg (SD: 2.83) in non-progressors, and the median follow-up IOP increased from 23.37 mmHg in non-progressors to 25.06 mmHg in progressors. Similarly, the maximum recorded IOP was higher in the progression group, with a mean of 30.02 mmHg (SD: 2.95) compared to 28.31 mmHg (SD: 2.85) in non-progressors. The variability of IOP, as measured by standard deviation (SD) and range, also differed between the groups. The mean IOP SD was 2.65 mmHg (SD: 0.96) in the progression group and 2.39 mmHg (SD: 0.94) in the non-progression group, while the mean IOP range was wider in progressors (9.14 mmHg) compared to non-progressors (8.35 mmHg). These findings suggest that greater IOP variability may be associated with an increased risk of POAG progression. Corneal thickness, measured as a proxy for central corneal thickness, showed similar mean values across both groups (574.53 μ m for progressors vs. 574.50 μ m for non-progressors). The median corneal thickness also exhibited minimal variation between groups, indicating that this variable may not be a distinguishing factor in POAG progression within this cohort. The vertical cup-to-disc ratio (VCDR) was slightly higher in the progression group, with a mean of 0.41 (SD: 0.17) compared to 0.40 (SD: 0.18) in non-progressors. Median values further supported this finding, with progressors showing a higher VCDR of 0.42 compared to 0.40 in non-progressors, reinforcing the role of structural optic nerve changes in glaucoma development.

Table 2. Comparison of Mean (SD) and Median (IQR) for Continuous Variables Between Censored Patients and Those Who Progressed to POAG.

Variable	Mean (SD)	Mean (SD)	Median (IQR)	Median (IQR)
	Censored Group	Progression Group	Censored Group	Progression Group
Age (years)	55.50 (9.87)	55.49 (10.15)	55.85 (48.65–62.33)	55.77 (48.44–62.17)
Base IOP	24.32 (2.70)	25.48 (2.99)	24.40 (22.37–26.17)	25.41 (23.64–27.50)
Corneal Thickness (μm)	574.50 (39.60)	574.53 (36.64)	577.33 (546.47–602.28)	573.66 (550.24–600.03)
Vertical cup-to-disc ratio	0.40 (0.18)	0.41 (0.17)	0.40 (0.27–0.51)	0.42 (0.29–0.53)
¹ FUP IOP (mmHg)	23.21 (2.83)	25.02 (3.06)	23.37 (21.42–25.13)	25.06 (23.07–27.13)
² SD IOP (mmHg)	2.39 (0.94)	2.65 (0.96)	2.39 (1.75–2.98)	2.66 (1.90–3.32)
³ Max IOP (mmHg)	28.31 (2.85)	30.02 (2.95)	28.05 (26.32–30.28)	29.68 (27.72–32.05)
⁴ Range IOP (mmHg)	8.35 (3.14)	9.14 (3.15)	8.33 (6.33–10.33)	8.99 (6.83–11.33)
⁵ CV IOP (mmHg)	0.10 (0.04)	0.11 (0.04)	0.10 (0.07–0.13)	0.10 (0.08–0.13)

¹FUP IOP: Mean intraocular pressure measured during the follow-up period, ²SD IOP: Standard deviation of intraocular pressure during the follow-up period, ³Max IOP: Maximum intraocular pressure recorded during the follow-up period, ⁴Range IOP: Difference between the highest and lowest intraocular pressure values during the follow-up period, ⁵CV IOP: Coefficient of variation for intraocular pressure, calculated as the standard deviation divided by the mean IOP.

To evaluate potential multicollinearity among IOP metrics, a correlation matrix was constructed, focusing specifically on the relationships between Base IOP, FUP IOP, SD IOP, Max IOP, Range IOP, and CV IOP (Figure 1). The analysis revealed a strong positive correlation between SD IOP and CV IOP (correlation coefficient = 0.93), indicating significant redundancy between these two measures of IOP variability. Additionally, a moderate negative correlation was observed between CV IOP and FUP IOP (correlation coefficient = -0.32), while weaker correlations were identified between other IOP metrics, suggesting minimal overlap among them.

Based on these findings, CV IOP was excluded from further analysis to mitigate multicollinearity and ensure model stability. This decision allowed the retention of SD IOP, which provides meaningful insights into IOP variability, and other key metrics, including base IOP, FUP IOP, Range IOP, and MAX IOP, which represent different aspects of IOP dynamics. By excluding highly correlated variables, the remaining metrics were selected to contribute independently to the Cox proportional hazards model, enhancing the model's interpretability and reliability in assessing POAG progression risk.

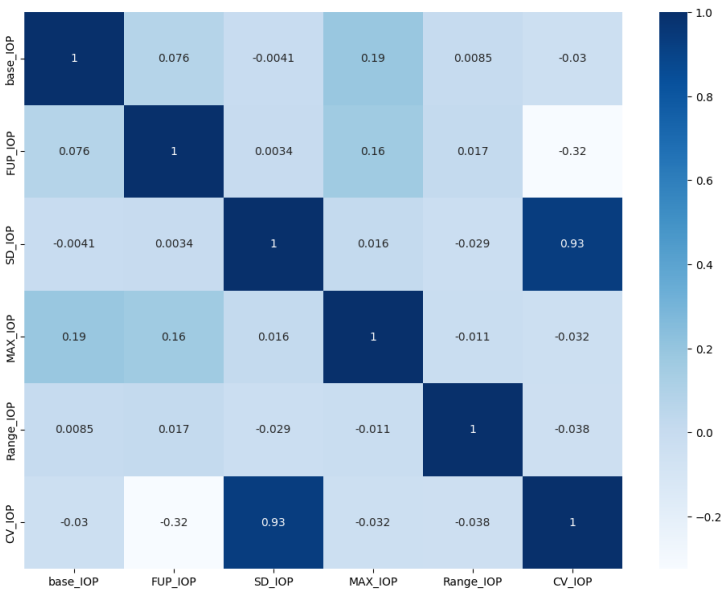


Figure 1. Correlation Matrix of IOP Metrics to Assess Multicollinearity Among Variables

3.2 Kaplan-Meier Survival Analysis

The Kaplan-Meier survival curve presented in Figure 2a shows the time to progression to POAG among the study cohort. The median time to progression was 3644 days, with a 95% confidence interval ranging from 2734 days to infinity. The survival probability declines rapidly during the first 500 days, indicating an early phase where progression is more common, and then decreases more gradually over the remainder of the follow-up period. The number of participants at risk decreases steadily, starting from 819 at baseline and dropping to 487 at 500 days, 135 at 2500 days, and 78 at 3500 days. This reduction is accompanied by an increase in the number of censored participants, reflecting the longitudinal nature of the study. The confidence bands around the survival estimates widen over time, highlighting increased uncertainty in the survival probabilities as fewer participants remain at risk in the later stages.

3.3 Nelson-Aalen estimation cumulative hazard function

The Nelson-Aalen cumulative hazard estimation curve presented in Figure 2b provides a complementary perspective to the Kaplan-Meier survival curve shown in Figure 1a. The increasing cumulative hazard over time reflects the gradual accumulation of risk for progression to POAG in the study cohort. Like the survival analysis, the early phase of follow-up demonstrates a steeper rise in hazard, highlighting the higher risk of progression during this period, while the subsequent increase is more gradual.

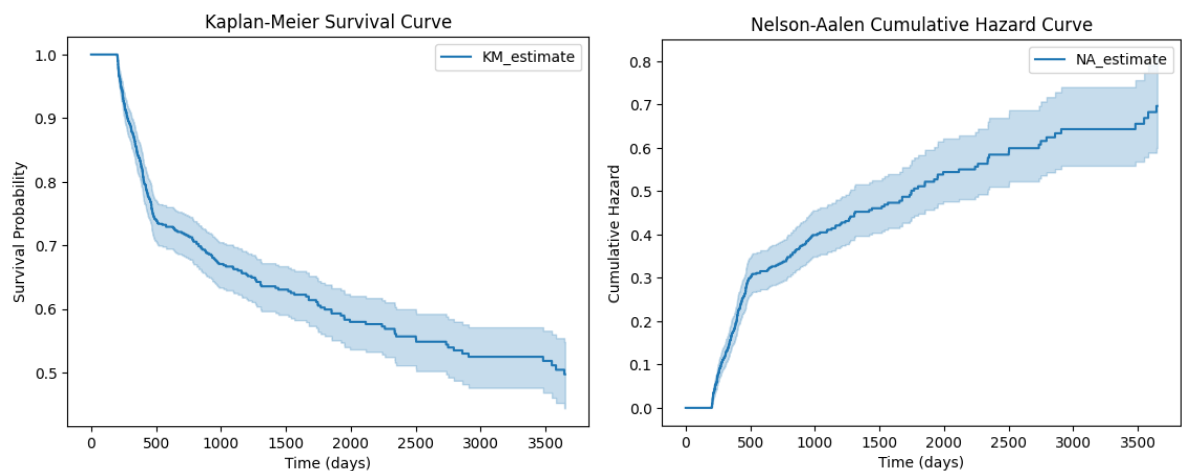


Figure 2. (a) Kaplan-Meier Survival Curve for Time to POAG Progression (b) Nelson-Aalen Cumulative Hazard Function for POAG Progression

3.4 Log-Rank Test Results

To assess whether survival times (time to progression to POAG) differed significantly across groups for the categorical variables in the dataset, log-rank tests were conducted for each variable.

Among the categorical variables analyzed, hypertension emerged as the only variable with a statistically significant difference in survival distributions. The log-rank test for hypertension produced a highly significant p-value (<0.005), indicating that participants with hypertension experienced different survival probabilities compared to those without hypertension. The Kaplan-Meier survival curves in Figure 3 visually reinforce the role of hypertension as a potential risk factor. Participants with hypertension consistently show lower survival probabilities compared to those without hypertension throughout the follow-up period. This aligns with the statistical findings and highlights the potential impact of hypertension on the progression to POAG.

Conversely, no significant differences in survival distributions were observed for other variables, including sex ($p=0.29$), family history of glaucoma ($p=0.41$), diabetes ($p=0.12$), cardiovascular disease (CVD) ($p=0.40$), migraine ($p=0.73$), and ethnicity ($p=0.77$).

These results emphasize the importance of hypertension as a potential driver of faster disease progression while suggesting that the other categorical variables analyzed may not independently influence POAG risk in this cohort.

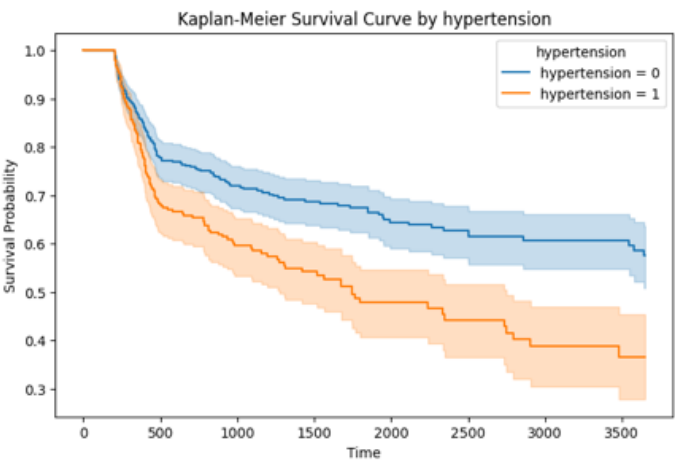


Figure 3. Kaplan-Meier Survival Curve Stratified by Hypertension Status

3.5 Cox Proportional Hazards Model for Open-Angle Glaucoma

In the first iteration, all available variables, except CV IOP, were included in the Cox model. This comprehensive model achieved a mean concordance index of 0.7415 (Standard deviation: 0.0503) using 10-fold cross-validation. While the model provided reasonable predictive performance, it was complex and included variables that may not contribute significantly to prediction or could violate modeling assumptions. To address this, Forward selection and Backward selection were applied as intermediate steps for variable selection. The covariates included in these selection processes were all numerical variables and the categorical variable hypertension, as the log-rank test had shown it to be significant. The remaining categorical variables were excluded due to their high p-values in the initial model and non-significant results in the log-rank test. Both Forward and Backward selection methods identified the same set of variables: Hypertension, Sex, Range IOP, Max IOP, SD IOP, FUP IOP, VCDR, Corneal thickness, Base IOP, Age.

A Cox model was then constructed using these variables, achieving a concordance index of 0.7451 (Standard deviation: 0.0483). However, this model still included variables with non-significant p-values, prompting the use of Lasso penalization as an additional step for variable selection. The penalization technique identified only the most meaningful predictors by shrinking the coefficients of less relevant variables toward zero. The penalized model maintained the same concordance index of 0.7415 (Standard deviation: 0.0477), demonstrating that the model's predictive performance remained consistent despite the reduction in complexity and the selection of only the most relevant variables, thereby improving its interpretability.

The variables with non-zero coefficients after penalization were:

- Base IOP: Hazard ratio = 1.22 (95% CI: 1.08–1.37, $p < 0.005$)
- FUP IOP: Hazard ratio = 1.47 (95% CI: 1.30–1.67, $p < 0.005$)
- Max IOP: Hazard ratio = 1.43 (95% CI: 1.26–1.61, $p < 0.005$)
- Range IOP: Hazard ratio = 1.13 (95% CI: 1.00–1.28, $p < 0.005$)
- SD IOP: Hazard ratio = 1.16 (95% CI: 1.03–1.31, $p < 0.005$)
- Hypertension: Hazard ratio = 1.24 (95% CI: 0.97–1.57, $p = 0.08$)

To ensure the model's clinical applicability and further simplify it, these variables were retained in the final version of the model, as the concordance index remained consistent with previous iterations (0.7457, standard deviation: 0.0449). Additionally, the partial log-likelihood of the model containing only the key variables was -1594.55, compared to -1667.59 for the model with penalizations, further indicating the improved balance between simplicity and model fit. This refinement enhances the model's usability while preserving its predictive accuracy, making it more suitable for integration into clinical decision-making. The next step was to verify compliance with the proportional hazards assumption, which revealed that hypertension violated this requirement.

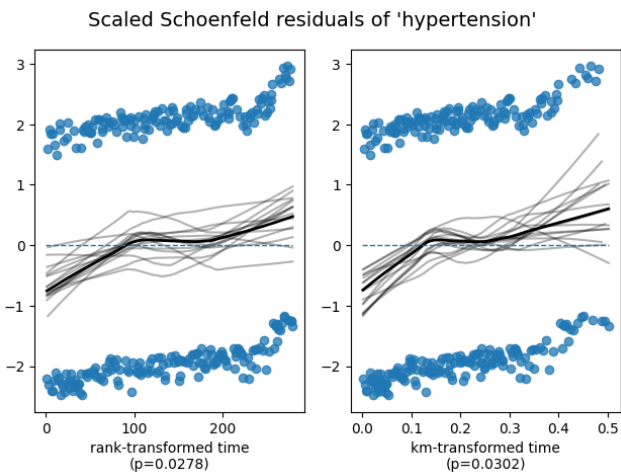


Figure 4. Scaled Schoenfeld Residuals for Hypertension Highlighting Violation of the Proportional Hazards Assumption

Analysis of Schoenfeld residuals, as shown in Figure 4, indicates a time-dependent effect: initially, the residuals are negative, reflecting an overestimation of POAG progression risk in hypertensive individuals, but they shift to positive values over time, suggesting underestimation. Consequently, hypertension was removed as a covariate and instead included as a stratification variable in the final model to address this temporal inconsistency and ensure the model's validity. The coefficients and hazard ratios (HRs) derived from the Cox proportional hazards model, as depicted in Figure 5, illustrate the relative contribution of each variable to the risk of progression to POAG. Baseline IOP emerged as a significant predictor, with a hazard ratio of 1.41 (95% CI: 1.26–1.58, $p < 0.005$), indicating that each mmHg increase in baseline IOP corresponds to a 41% higher risk of progression. Similarly, the mean IOP during follow-up (FUP IOP) was identified as the strongest predictor, with a hazard ratio of 1.75 (95% CI: 1.54–1.98, $p < 0.005$), signifying a 75% increase in risk per unit increase. Maximum IOP (Max IOP) during the follow-up period also demonstrated a strong association with progression risk, with a hazard ratio of 1.60 (95% CI: 1.41–1.81, $p < 0.005$), suggesting that peaks in IOP substantially elevate the likelihood of disease advancement.

Metrics assessing IOP variability were equally impactful. The standard deviation of IOP (SD IOP) had a hazard ratio of 1.35 (95% CI: 1.19–1.52, $p < 0.005$), while the range of IOP (Range IOP) showed a hazard ratio of 1.40 (95% CI: 1.24–1.58, $p < 0.005$). Hypertension was used as a stratification variable, allowing the baseline hazard to vary between participants with and without hypertension. The results of the final model are summarized below:

- Mean Concordance Index: 0.7370 (Standard deviation: 0.0523)
- Partial AIC: 2809.18
- Log-likelihood ratio test: 234.11 on 5 degrees of freedom (p -value < 0.005)
- Partial log-likelihood: -1399.59

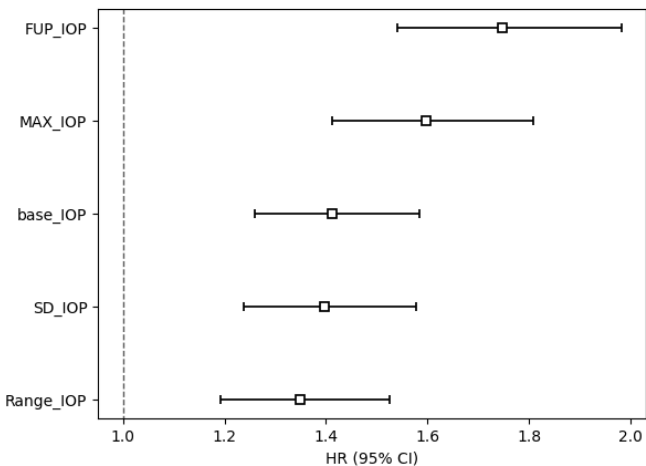


Figure 5. Hazard Ratios (HR) and 95% Confidence Intervals for Key Predictors of POAG Progression in the Final Cox Model

The final model was tested for compliance with the proportional hazards assumption using Schoenfeld residuals, which confirmed that all included variables satisfied the assumption. Additionally, Martingale residuals and deviance residuals were examined to evaluate model fit and identify outliers. Both sets of residuals showed no significant deviations or outliers, confirming the adequacy of the model:

- Martingale residuals: All values close to 0, indicating a good model fit.
- Deviance residuals: All values close to 0, confirming no evidence of poor fit or influential observations.

This final model represents a balance between predictive performance, interpretability, and compliance with modeling assumptions, providing a robust tool for understanding the factors influencing the progression to POAG.

3.6 Analysis of IOP Variability across Patient Subgroups

Building upon the findings of the Cox proportional hazards model, which identified SD IOP and Range IOP as significant predictors of progression to POAG, further analysis using the Log-Rank test was conducted to explore whether the impact of IOP variability differs across patient subgroups. This analysis revealed significant differences in progression risk, highlighting the interaction between IOP variability metrics and systemic or demographic factors (Table 3). The results demonstrated that SD IOP significantly influenced progression risk in males ($p = 0.0205$), while Range IOP had a stronger impact in females ($p = 0.0095$), suggesting sex-specific differences in how cumulative fluctuations versus extreme peaks in IOP contribute to disease progression. Patients without a family history of glaucoma showed significant associations for both SD IOP ($p = 0.0043$) and Range IOP ($p = 0.0017$), whereas no significant effects were observed in those with a family history. These findings suggest that IOP variability may be particularly relevant in patients whose risk is not predominantly driven by genetic predisposition.

Among systemic factors, diabetes amplified the effect of both variability metrics, with diabetic patients exhibiting stronger associations (SD IOP: $p = 0.0059$; Range IOP: $p = 0.0202$) compared to non-diabetics (SD IOP: $p = 0.0360$; Range IOP: $p = 0.0145$). Similarly, hypertensive patients showed a greater sensitivity to Range IOP ($p = 0.0085$), while SD IOP was more impactful in non-hypertensive individuals ($p = 0.0065$), reflecting potential differences in how systemic blood pressure regulation modulates IOP variability effects. For cardiovascular disease (CVD), both SD IOP and Range IOP were significant only in patients without CVD ($p < 0.005$ for both metrics), indicating that cardiovascular factors may mitigate the impact of IOP variability on glaucoma progression.

Interestingly, patients without migraines showed significant associations with both SD IOP ($p = 0.0040$) and Range IOP ($p = 0.0053$), while no such effects were observed in patients with migraines.

These findings underscore the heterogeneity in how IOP variability contributes to glaucoma progression, with distinct effects observed across subgroups.

Table 3: Subgroup Analysis of the Interaction Between IOP Variability Metrics (SD IOP and Range IOP) and Systemic or Demographic Factors in POAG Progression Risk

Variable (Categories)	P-value for SD IOP (High vs Low)	P-value for Range IOP (High vs Low)
Sex (Male, Female)	0.0205 (Male), 0.0581 (Female)	0.0600 (Male), 0.0095 (Female)
Family History (No, Yes)	0.0043 (No), 0.2652 (Yes)	0.0017 (No), 0.2487 (Yes)
Diabetes (No, Yes)	0.0360 (No), 0.0059 (Yes)	0.0145 (No), 0.0202 (Yes)
Hypertension (No, Yes)	0.0065 (No), 0.0917 (Yes)	0.0680 (No), 0.0085 (Yes)
Cardiovascular Disease (No, Yes)	0.0008 (No), 0.4115 (Yes)	0.0010 (No), 0.5603 (Yes)
Migraine (No, Yes)	0.0040 (No), 0.5053 (Yes)	0.0053 (No), 0.1598 (Yes)

4. Discussion

This study aimed to identify key predictors of POAG progression in patients with elevated IOP by leveraging robust survival analysis methods, including a Cox proportional hazards model. The findings provide critical insights into the multifactorial nature of POAG progression, emphasizing the roles of IOP-related parameters and systemic factors in shaping disease risk.

Hypertension emerged as a significant predictor, reinforcing its potential role in glaucoma progression through vascular regulation and optic nerve health. However, the complex and time-dependent effects of hypertension, as evidenced by its violation of the proportional hazards assumption, necessitated its inclusion as a stratification variable. Despite

this adjustment, hypertension remained an important factor, with log-rank tests indicating a higher probability of progression in hypertensive patients compared to non-hypertensive individuals. These results align with some existing studies but also contribute to the ongoing debate regarding the relationship between systemic hypertension and glaucoma risk, particularly highlighting the need for further exploration of vascular mechanisms.

In addition to systemic factors, the study highlights the importance of IOP metrics in predicting POAG progression. Baseline IOP and mean follow-up IOP were particularly strong predictors, corroborating findings from prior studies such as the Early Manifest Glaucoma Trial (EMGT). Maximum IOP also emerged as a key contributor, underscoring the importance of addressing peak pressures in disease management. Furthermore, IOP variability, represented by standard deviation (SD IOP) and range (Range IOP), was independently associated with progression. These results align with studies like the Advanced Glaucoma Intervention Study (AGIS), which linked greater IOP fluctuations with faster visual field loss, and highlight the need to address IOP variability in therapeutic strategies alongside sustained IOP control.

Building on the Cox model, subgroup analyses revealed important nuances in how IOP variability affects different patient populations. For instance, SD IOP had a more pronounced impact in males and non-hypertensive patients, while Range IOP was more relevant for females and hypertensive individuals. These distinctions underscore the interplay between demographic, systemic, and ocular factors in shaping disease progression, reinforcing the need for personalized approaches to glaucoma management. The study also identified variables, such as age, sex, ethnicity, diabetes, corneal thickness, and vertical cup-to-disc ratio (VCDR), that were not significant predictors of POAG progression in this model. These findings suggest that their roles may be more relevant to disease onset rather than progression, or that their influence is mediated through other factors such as IOP.

The findings of this study have significant implications for clinical practice and public health. By incorporating IOP variability metrics into clinical decision-making, ophthalmologists can improve risk stratification and develop tailored treatment strategies for high-risk patients. For instance, patients with elevated IOP variability may benefit from therapies aimed at stabilizing fluctuations, while those with persistently elevated maximum IOP might require more aggressive IOP-lowering interventions. The subgroup-specific insights further support the need for personalized care, as management strategies can be tailored based on demographic and systemic characteristics, such as sex, hypertension, and diabetes. From a public health perspective, the results emphasize the importance of consistent and accessible IOP monitoring as a cornerstone of glaucoma management. Regular assessments of both absolute IOP levels and variability can facilitate early detection and intervention, particularly in patients with systemic comorbidities. These findings also underscore the need for integrated care approaches that address both ocular and systemic health, providing a framework for more holistic management strategies.

5. Conclusion

This study contributes to a deeper understanding of the multifactorial nature of POAG progression, highlighting the roles of IOP variability and systemic factors like hypertension. Future research should focus on validating these findings in larger and more diverse populations, as well as exploring the mechanisms underlying the interaction between systemic health and IOP dynamics. These insights offer a pathway for refining clinical guidelines and improving outcomes for patients at risk of POAG through more personalized and targeted approaches.

References

1. Collaborative Normal-Tension Glaucoma Study Group. The effectiveness of intraocular pressure reduction in the treatment of normal-tension glaucoma. *Am J Ophthalmol* 1998, 126(4): 498–505. doi:10.1016/S0002-9394(98)00108-0.
2. Heijl, A.; Leske, M.C.; Bengtsson, B.; et al. Reduction of intraocular pressure and glaucoma progression: Results from the Early Manifest Glaucoma Trial. *Arch Ophthalmol* 2002, 120(10): 1268–1279. doi:10.1001/archopht.120.10.1268.
3. Dielemans, I.; Vingerling, J.R.; Algra, D.; et al. Primary open-angle glaucoma, intraocular pressure, and systemic blood pressure in the general elderly population: The Rotterdam Study. *Ophthalmology* 1995, 102(1): 54–60. doi:10.1016/S0161-6420(95)31054-8.